

Individual Reflection on Group Project

Project Title: Edge-Deployable Rock Detection and Burial Status Classification

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Team / Group: Team 9

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1. Your views on the completing the project as a team

Collaborating as a team was beneficial due to the project including data preparation, model development, edge deployment on the Raspberry Pi 5, visualisation, and academic reporting. My tasks involved dataset preparation and splitting, image augmentation, YOLO augmentation control and training configuration, YOLOv8n and YOLO11n training, evaluation and Tableau visualisations for the final report. These tasks were divided to make progress in parallel through shared datasets, checkpoints and metrics. Since my outputs were to be used in subsequent analyses, an incorrect result could have affected the whole team. I therefore found that the importance of sharing traceable outcomes and limitations was at least as great as the importance of training the models.

2. Your views on project management and execution of your team

There were definite dependency orderings among our executions. We worked with an original dataset of 448 images containing 706 annotated rocks, then created the 313/45/90 train-validation-test split before augmentation. I created six photometric variants only for training, increasing that partition to 2,191 images while leaving the validation and test sets unchanged. Common YOLO training configurations made the model comparisons meaningful, and Tableau visuals expressed the evidence effectively. I learned that efficient management involves having criteria for evaluation, uniformity of naming conventions for files, and early confirmation of authoritative results when multiple collaborators were working on the project.

3. Reflections on Successes, Failures, and Retrospective Changes

- **What Worked Well:** Splitting prior to augmentation stopped data leakage, and a shared configuration was used to make the comparison fair. Based on the predefined validation mAP_{50–95} metric, YOLOv8n was selected with a score of 0.4288, although other models led on other metrics. Tableau clarified these tradeoffs.
- **What Did Not Work Well:** The smaller dataset and the difficulty of identifying the half-buried class made generalisation challenging. The YOLOv8n model's performance decreased from 0.4288 mAP_{50–95} in the validation set to 0.3384 in the test set. It was important to control the offline and internal YOLO augmentation properly to avoid an overly aggressive pipeline.
- **What I Would Do Differently:** I would define the experiment matrix and visualisation structure earlier and, if time permitted, carry out an augmentation ablation test. Earlier joint reviews would also ensure that there is consistency in metrics, labels and reporting terminology.

4. Any other reflections

This project helped me improve in terms of taking ownership of a full experimental process chain and at the same time contributing to a joint research result. I became more aware of data leakage, reproducibility and the distinction between choosing a model based on some criterion and saying that this is the best model. The most important lesson for me was that thorough research does not mean hiding shortcomings, such as the half-buried recognition gap, or presenting only positive outcomes.